

2017

COMPUTER SCIENCE AND ENGINEERING

Paper – CSCL-0901

(Topics in Algorithms)

Full Marks – 70

The figures in the margin indicate full marks

Candidates are required to give their answers in their own words as far as practicable

Group – A

Answer **any five** questions

2×5

1. Solve the following recurrence equation by Master Theorem.

$$T(n) = 3T(n/2) + n^2.$$

2. What is meant by Pspace?
3. State Fermat's Little Theorem. What are Pseudo-primes?
4. What is meant by Decision problem? Give an example.
5. State base-function property of the family of algorithms.
6. What is meant by SM SIMD computers?
7. What is meant by Odd-even merging network?

Group – B

Answer **any five** questions

8. State and explain Master's Theorem. 1+3
9. Write a brief note on Red-Black Trees. 4
10. What are online algorithms? What do you mean by competitive analysis in regard to online algorithms? 2+2
11. Write a randomized algorithm for computing the value of π . Explain the steps. 4
12. Define primitive recursion. Show how two-variable addition function can be defined by primitive recursion. 1+3
13. Given a flow F in a network, prove that the flow out of the 'source' a equals the flow into the sink z . 4
14. Suppose that α is a family of algorithms in which properties 1, 2 and 4 are satisfied. Then show that the one-variable function f where

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x \neq 0 \end{cases}$$

is algorithmic in α .

4

[Turn Over]

Group - C

Answer **any four** questions

15. State and establish Chinese Remainder Theorem. 2+8
16. Explain what is meant by convex hull. Write Graham's Algorithm for computing the convex hull. Illustrate. 2+8
17. Explain, with an example, how does KMP algorithm work. 10
18. Prove that 3SAT problem is NP-complete. 10
19. Describe the twice-around-the-tree algorithm for Travelling Salesperson problem. Prove that it is a 2-approximaion algorithm with Euclidean distances. 5+5
20. Write a divide and conquer algorithm for finding the distance between a closest pair of points. 10
21. State the essential features of a parallel searching algorithm based on EREW architecture. 10

Group - B

Answer **any five** questions

8. State and explain Master's Theorem. 1+3
9. Write a brief note on Red-Black Trees. 4
10. What are online algorithms? What do you mean by competitive analysis in regard to online algorithms? 2+3
11. Write a randomized algorithm for computing the value of π . Explain the steps. 4
12. Define primitive recursion. Show how two-variable addition function can be defined by primitive recursion. 1+3
13. Given a flow F in a network, prove that the flow out of the source 's' equals the flow into the sink 't'. 4
14. Suppose that α is a family of algorithms in which properties 1, 2 and 4 are satisfied. Then show that the one-variable function f where

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x \neq 0 \end{cases}$$

is algorithmic in α .

[Turn Over]